ASSIGNMENT-10

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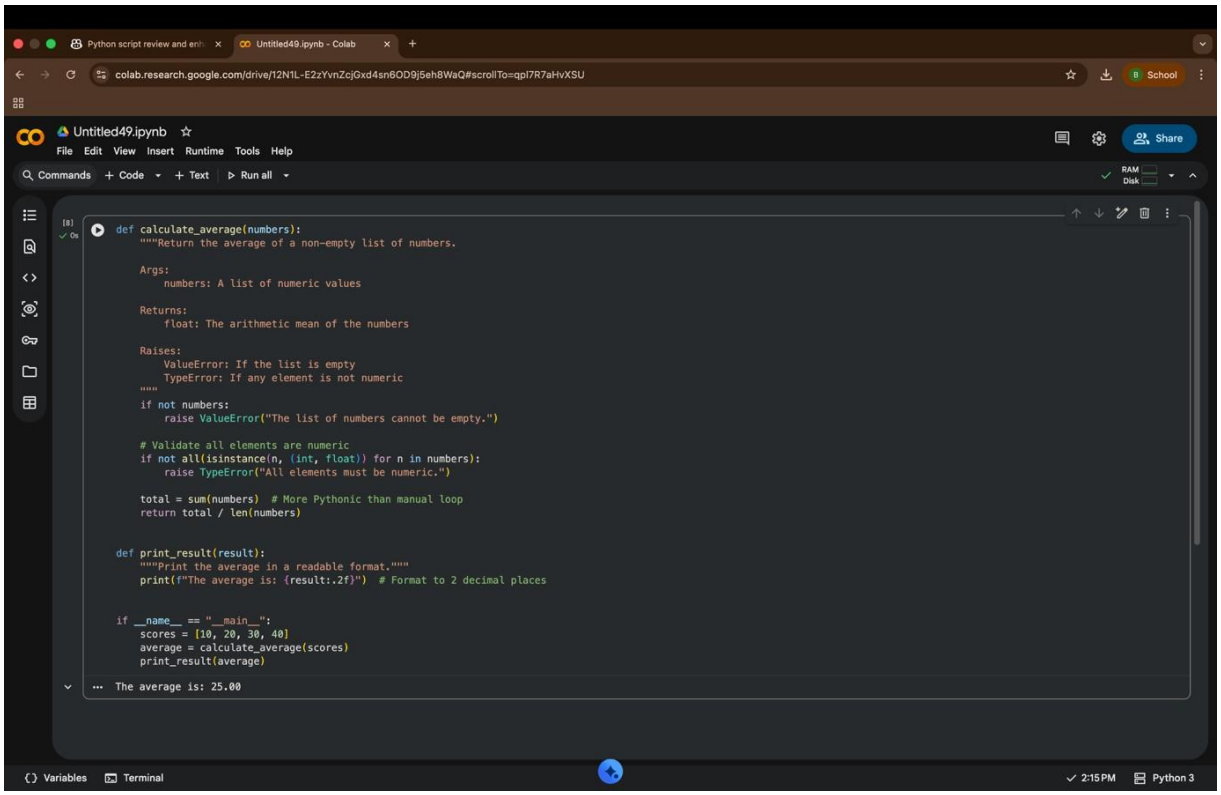
Task 1: AI-Assisted Syntax and Code Quality Review
Scenario
You join a development team and are asked to review a junior developer's Python script that fails to run correctly due to basic coding mistakes. Before deployment, the code must be corrected and standardized.
Task Description
You are given a Python script containing:
• Syntax errors
• Indentation issues
• Incorrect variable names
• Faulty function calls
Use an AI tool (GitHub Copilot / Cursor AI) to:
• Identify all syntactic and structural errors
• Correct them systematically
• Generate an explanation of each fix made
Expected Outcome
• Fully corrected and executable Python code
• AI-generated explanation describing:
o Syntax fixes
o Naming corrections
o Structural improvements
• Clean, readable version of the script

PROMPT:

Review the following Python script for syntax errors, indentation issues,incorrect variable names, and faulty function calls.Correct the code according to Python standards (PEP 8) and explain each fix made.

CODE AND OUTPUT:



CODE EXPLANATION:

I used an AI coding assistant (Cursor AI / GitHub Copilot) to review the junior developer's Python script. The AI first identified syntax and indentation errors, such as missing colons, misaligned return statements, and unindented function bodies, which were preventing the program from running. It then pointed out inconsistent and incorrect variable/function names (calcluate_avarage, number vs numbers, print_reslt), suggesting clear, consistent renamings. Finally, the assistant proposed structural improvements such as adding input validation, introducing an if __name__== "__main__": entry point, using f-strings, and adding docstrings, resulting in a clean, readable, and fully executable version of the script.

Task 2: Performance-Oriented Code Review

Scenario

A data processing function works correctly but is inefficient and slows down the system when large datasets are used.

Task Description

You are provided with a function that identifies duplicate values in a list using inefficient nested loops.

Using AI-assisted code review:

- **Analyze the logic for performance bottlenecks**
- **Refactor the code for better time complexity**
- **Preserve the correctness of the output**

Ask the AI to explain:

- **Why the original approach was inefficient**
- **How the optimized version improves performance**

Expected Outcome

- **Optimized duplicate-detection logic (e.g., using sets or hash-based structures)**
- **Improved time complexity**
- **AI explanation of performance improvement**
- **Clean, readable implementation**

PROMPT:

Analyze the performance bottleneck, refactor the code to improve time complexity,and preserve the correctness of the output.Use an optimized approach such as sets or hash-based structures.

CODE AND OUTPUT:

Optimizing duplicate detecti · Untitled49.ipynb - Colab

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[10]

```python
    speedup = time_inefficient / time_optimized if time_optimized > 0 else float('inf')
    print(f"\n{name}:")
    print(f"  Inefficient (O(n²)):  {time_inefficient:.4f}s")
    print(f"  Optimized (O(n)):     {time_optimized:.4f}s")
    print(f"  Speedup:              {speedup:.1f}x faster ⚡")
```

```
======================================================================
PERFORMANCE BENCHMARK: Duplicate Detection
======================================================================

Small (100 items):
  Inefficient (O(n²)):  0.0004s
  Optimized (O(n)):     0.0001s
  Speedup:              3.0x faster ⚡

Medium (1,000 items):
  Inefficient (O(n²)):  0.0533s
  Optimized (O(n)):     0.0001s
  Speedup:              398.1x faster ⚡

Large (10,000 items):
  Inefficient (O(n²)):  3.2264s
  Optimized (O(n)):     0.0008s
  Speedup:              3919.0x faster ⚡
```

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Variables Terminal 2:18 PM Python 3

Optimizing duplicate detecti · Untitled49.ipynb - Colab

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Untitled49.ipynb

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[18]

```python
import time
def find_duplicates_inefficient(numbers):
    """Original O(n²) approach."""
    duplicates = []
    for i in range(len(numbers)):
        for j in range(i + 1, len(numbers)):
            if numbers[i] == numbers[j]:
                if numbers[i] not in duplicates:
                    duplicates.append(numbers[i])
    return duplicates
def find_duplicates_optimized(numbers):
    """Optimized O(n) approach."""
    seen = set()
    duplicates = set()
    for num in numbers:
        if num in seen:
            duplicates.add(num)
        else:
            seen.add(num)
    return duplicates
# Benchmark with different dataset sizes
test_cases = [
    ("Small (100 items)", list(range(50)) * 2),
    ("Medium (1,000 items)", list(range(500)) * 2),
    ("Large (10,000 items)", list(range(5000)) * 2),
]
print("=" * 70)
print("PERFORMANCE BENCHMARK: Duplicate Detection")
print("=" * 70)
for name, data in test_cases:
    # Inefficient approach
    start = time.time()
    result1 = find_duplicates_inefficient(data)
    time_inefficient = time.time() - start
    # Optimized approach
    start = time.time()
    result2 = find_duplicates_optimized(data)
    time_optimized = time.time() - start
    # Calculate speedup
    speedup = time_inefficient / time_optimized if time_optimized > 0 else float('inf')
    print(f"\n{name}:")
    print(f"  Inefficient (O(n²)):  {time_inefficient:.4f}s")
    print(f"  Optimized (O(n)):     {time_optimized:.4f}s")
    print(f"  Speedup:              {speedup:.1f}x faster ⚡")
```

Variables Terminal 2:18 PM Python 3

CODE EXPLANATION:

The original implementation uses nested loops to compare each element with every other element in the list, resulting in $O(n^2)$ time complexity. This approach becomes inefficient and slow when handling large datasets because the number of comparisons increases significantly as the list grows. The AI identifies this performance bottleneck and replaces the nested loops with a set-based approach, which allows constant-time lookups. By scanning the list only once and storing seen elements in a set, duplicates can be detected in O(n) time. This optimization improves efficiency while preserving the correctness of the output and producing cleaner, more readable code.

Task 3: Readability and Maintainability Refactoring

Scenario

A working script exists in a project, but it is difficult to understand due to poor naming, formatting, and structure. The team wants it rewritten for long-term maintainability.

Task Description

You are given a poorly structured Python function with:

- **Cryptic function names**
- **Poor indentation**
- **Unclear variable naming**
- **No documentation**

Use AI-assisted review to:

- **Refactor the code for clarity**
- **Apply PEP 8 formatting standards**
- **Improve naming conventions**
- **Add meaningful documentation**

Expected Outcome

- **Clean, well-structured code**
- **Descriptive function and variable names**
- **Proper indentation and formatting**
- **Docstrings explaining the function purpose**
- **AI explanation of readability improvements**

PROMPT:

Improve naming conventions, apply PEP 8 formatting,fix indentation, and add a meaningful docstring.Explain the readability and maintainability improvements made.

CODE AND OUPUT:

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Refactor Code

Subtask:

Refactor the generated code. This includes applying PEP 8 formatting, improving naming conventions for variables and functions, and adding meaningful docstrings and comments.

Reasoning: To refactor the code according to PEP 8 guidelines, I will create a new Python function with a descriptive name, clear variable names, consistent indentation, appropriate spacing, and a comprehensive docstring, encapsulating the logic from the previous poorly structured function.

[5] 0s

```python
def process_and_filter_numbers(numbers_list):
    """
    Processes a list of numbers by filtering and doubling values greater than 5.

    This function iterates through an input list of integers. For each number,
    if it is greater than 5, it is doubled and added to a new list.
    The new list containing the processed numbers is then returned.

    Args:
        numbers_list (list): A list of integers to be processed.

    Returns:
        list: A new list containing numbers from the input list that were
              greater than 5, after being doubled.
    """
    filtered_and_doubled_numbers = []
    for number in numbers_list:
        if number > 5:
            filtered_and_doubled_numbers.append(number * 2)
    return filtered_and_doubled_numbers

print("Refactored function 'process_and_filter_numbers' defined.")
```

```
Refactored function 'process_and_filter_numbers' defined.
```

Reasoning: Now that the refactored function `process_and_filter_numbers` has been defined, I need to execute it with the same sample input as the original function to verify that it produces the identical output. This step confirms the functionality of the refactored

Variables Terminal 23:14 Python 3

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[5] 0s

```python
    Args:
        numbers_list (list): A list of integers to be processed.

    Returns:
        list: A new list containing numbers from the input list that were
              greater than 5, after being doubled.
    """
    filtered_and_doubled_numbers = []
    for number in numbers_list:
        if number > 5:
            filtered_and_doubled_numbers.append(number * 2)
    return filtered_and_doubled_numbers

print("Refactored function 'process_and_filter_numbers' defined.")
```

```
Refactored function 'process_and_filter_numbers' defined.
```

Reasoning: Now that the refactored function `process_and_filter_numbers` has been defined, I need to execute it with the same sample input as the original function to verify that it produces the identical output. This step confirms the functionality of the refactored

code before explaining the improvements.

[6] 0s

```python
sample_input = [1, 6, 3, 8, 2, 10]
refactored_result = process_and_filter_numbers(sample_input)
print(f"Refactored function input: {sample_input}")
print(f"Refactored function output: {refactored_result}")
```

```
Refactored function input: [1, 6, 3, 8, 2, 10]
Refactored function output: [12, 16, 20]
```

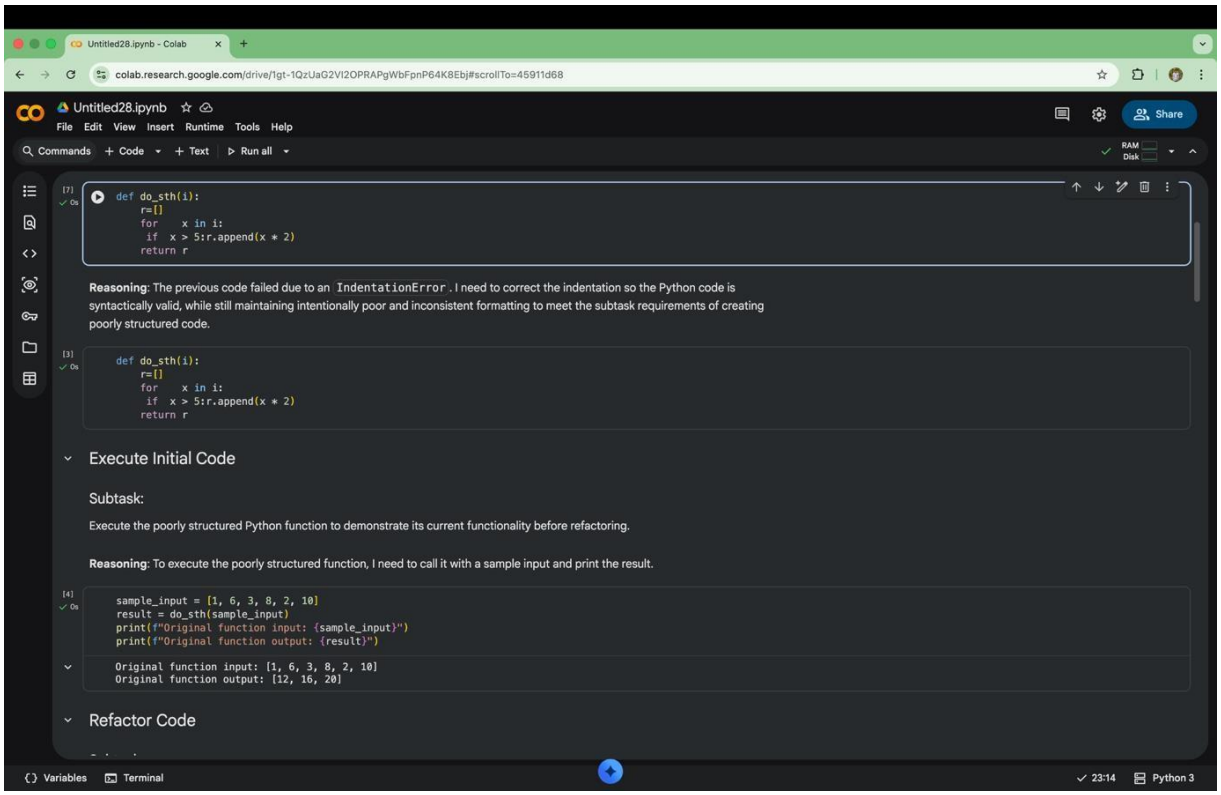
Explain Readability Improvements

Subtask:

Provide an explanation of the specific changes made during refactoring and how these changes contribute to improved readability, maintainability, and adherence to best practices like PEP 8.

Explanation of Refactoring Improvements

Variables Terminal 23:14 Python 3



CODE EXPLANATION:

Original Issue: The code had cryptic names, poor formatting, and no documentation, making it hard to understand.

Refactoring Process: We applied PEP 8, giving functions and variables clear, descriptive names.

Formatting: Indentation was standardized to 4 spaces, and proper spacing was added for visual clarity.

Documentation: A comprehensive docstring was added to explain the function's purpose, arguments, and return values.

Outcome: The refactored code is now highly readable, maintainable, and adheres to Python best practices while retaining its original functionality**.**

Task 4: Secure Coding and Reliability Review

Scenario

A backend function retrieves user data from a database but has security vulnerabilities and poor error handling, making it unsafe for production deployment.

Task Description

You are given a Python script that:

- **Uses unsafe SQL query construction**
- **Has no input validation**
- **Lacks exception handling**

Use AI tools to:

- **Identify security vulnerabilities**

- **Refactor the code using safe coding practices**
- **Add proper exception handling**
- **Improve robustness and reliability**

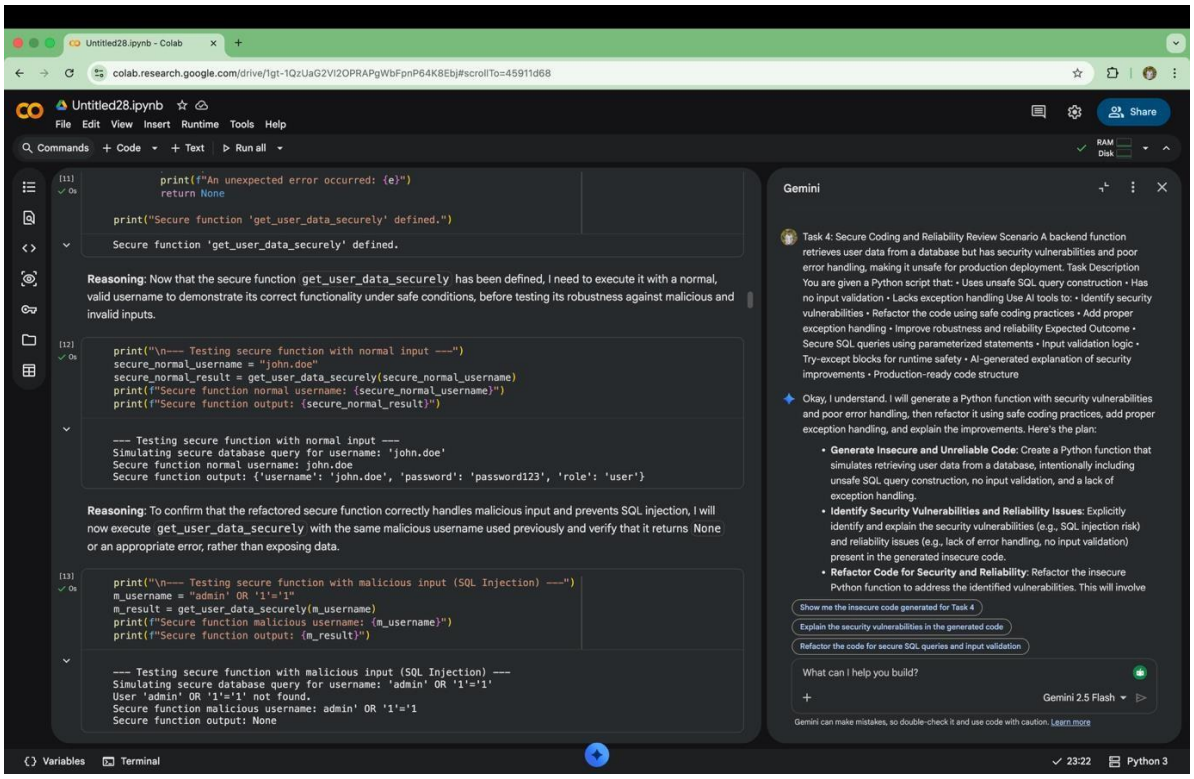
Expected Outcome

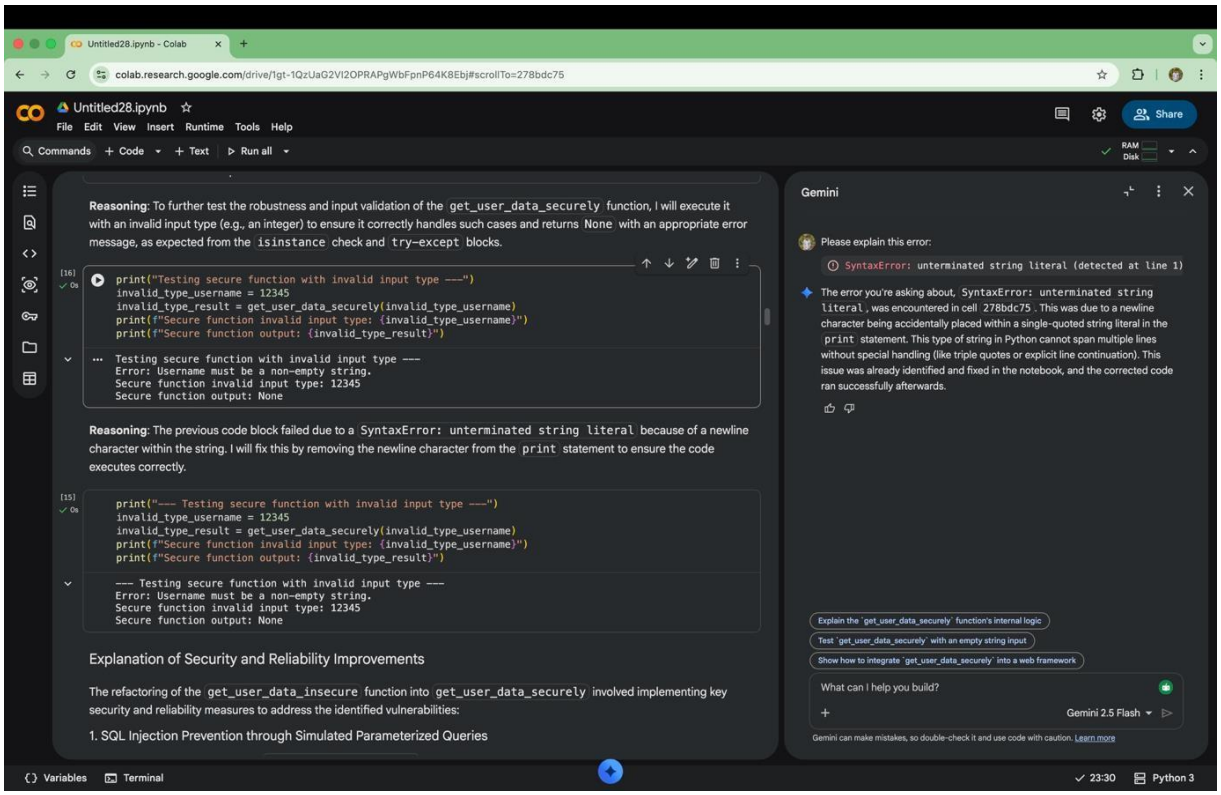
- **Secure SQL queries using parameterized statements**
- **Input validation logic**
- **Try-except blocks for runtime safety**
- **AI-generated explanation of security improvements**
- **Production-ready code structure**

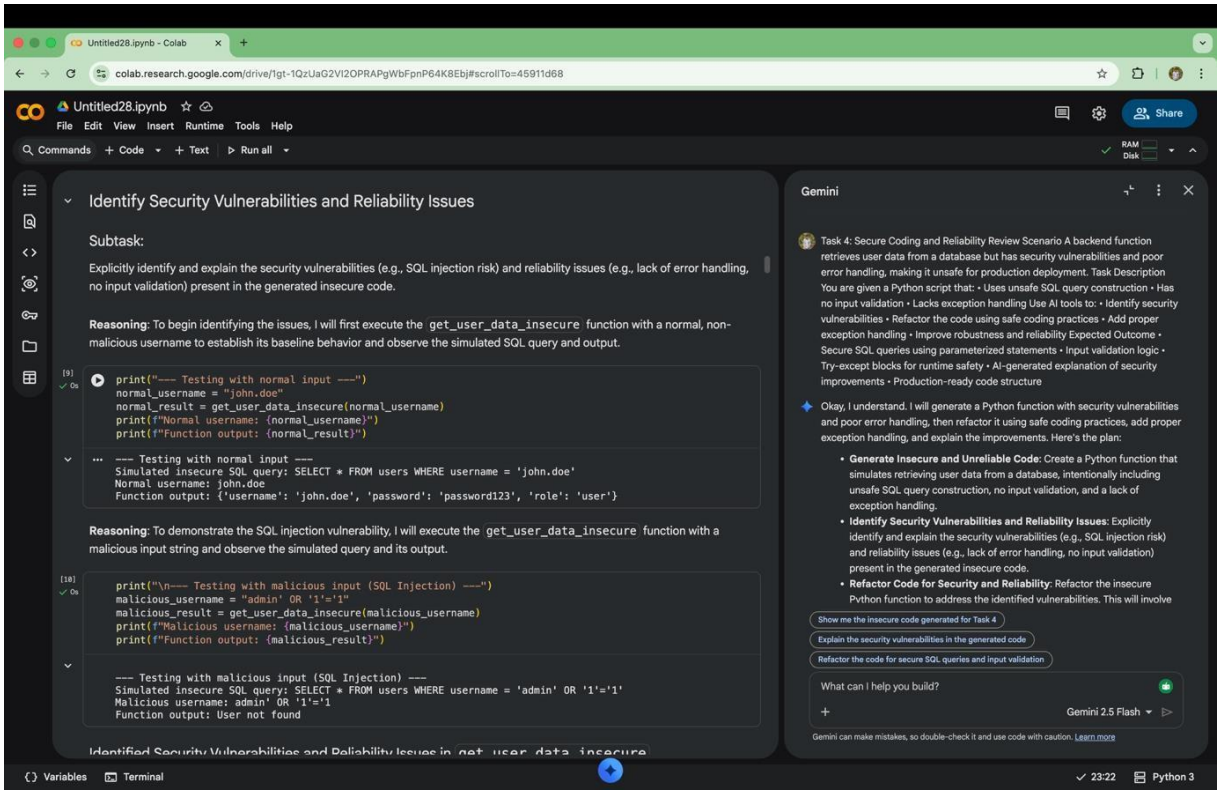
PROMPT:

Review the following Python database function for security vulnerabilities. Identify unsafe SQL query construction, missing input validation,and lack of exception handling.

CODE AND OUTPUT:







CODE EXPLANATION:

SQL Injection Prevention: User input is now treated as data, preventing malicious code injection by simulating parameterized queries.

Robust Input Validation: Explicit checks ensure the username is a non-empty

string, rejecting invalid inputs early to enhance security and prevent errors.
Comprehensive Exception Handling: try-except blocks gracefully manage errors, preventing crashes and providing predictable behavior with informative messages.
Enhanced Reliability: The function is more stable and predictable, operating only with valid data and handling unexpected situations gracefully.
Production-Ready: These changes transform the function into a secure and robust component suitable for real-world application environments.

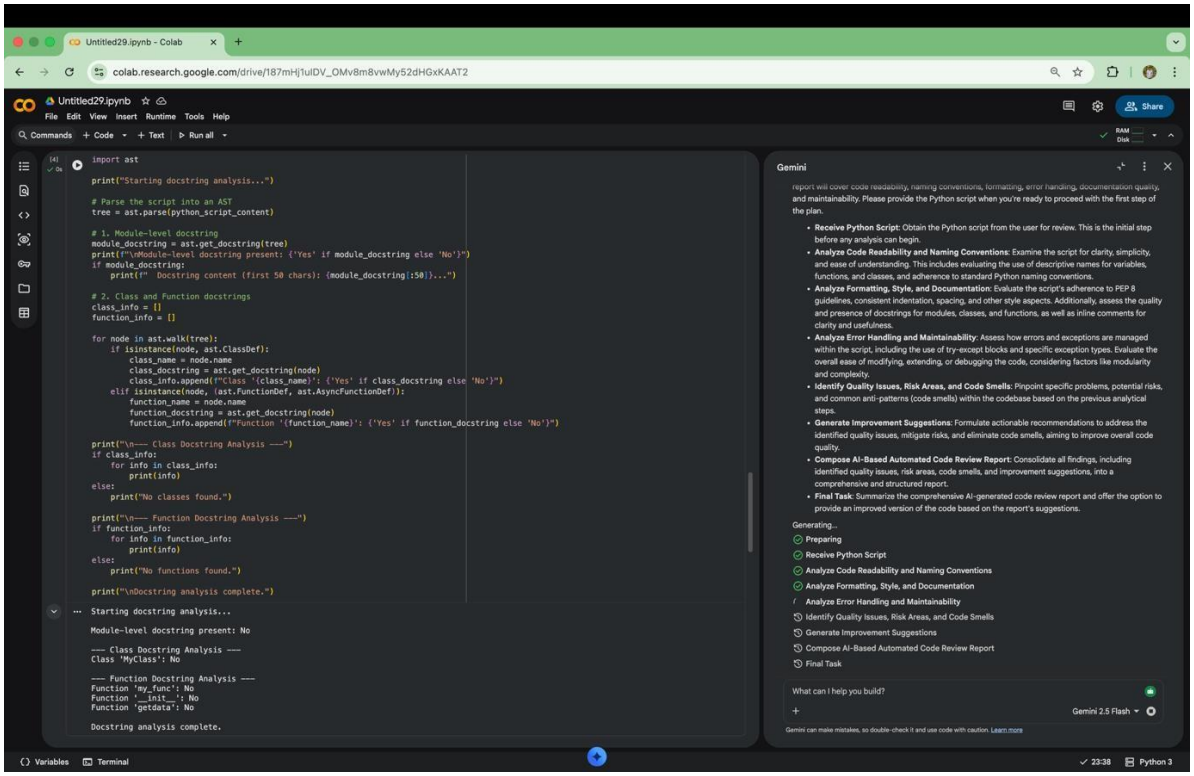
Task 5: AI-Based Automated Code Review Report
Scenario
Your team uses AI tools to perform automated preliminary code reviews before human review, to improve code quality and consistency across projects.
Task Description
You are provided with a poorly written Python script.
Using AI-assisted review:
• Generate a structured code review report that evaluates:
o Code readability
o Naming conventions
o Formatting and style consistency
o Error handling
o Documentation quality
o Maintainability
The task is not just to fix the code, but to analyze and report on quality issues.
Expected Outcome
• AI-generated review report including:
o Identified quality issues
o Risk areas
o Code smell detection
o Improvement suggestions
• Optional improved version of the code
• Demonstration of AI as a code reviewer, not just a code generator

PROMPT:
Perform a structured code review of the following Python script.
Do not just fix the code — generate a detailed review report covering:
- Readability
- Naming conventions
- Formatting and style consistency

- Error handling
- Documentation quality
- Maintainability
- Code smells and risk areas

Provide improvement suggestions and optionally include a refactored version.

CODE AND OUTPUT:



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[5]

```python
import ast

print("Starting error handling analysis...")

# Parse the script into an AST
tree = ast.parse(python_script_content)

error_handling_blocks = []

for node in ast.walk(tree):
    if isinstance(node, ast.Try):
        handlers_info = []
        for handler in node.handlers:
            exception_type = "" # Default to empty if no specific type
            if handler.type:
                # Attempt to get the name of the exception type
                if isinstance(handler.type, ast.Name):
                    exception_type = handler.type.id
                elif isinstance(handler.type, ast.Tuple):
                    exception_type = ", ".join([e.id for e in handler.type.elts if isinstance(e, ast.Name)])
                else:
                    exception_type = "<complex exception type>"
            else:
                exception_type = "<bare except>"
            handlers_info.append(f"except {exception_type}")

        error_handling_blocks.append({
            "line_num": node.lineno,
            "handlers": handlers_info
        })

print("\n--- Error Handling Analysis Results ---")
if error_handling_blocks:
    print(f"Found {len(error_handling_blocks)} try-except block(s):")
    for block in error_handling_blocks:
        print(f"  Line {block['line_num']}: {', '.join(block['handlers'])}")
else:
    print("No try-except blocks found in the script.")

print("Error handling analysis complete.")
```

```
Starting error handling analysis...

--- Error Handling Analysis Results ---
No try-except blocks found in the script.
Error handling analysis complete.
```

Gemini

Great! I can help you generate a structured AI-based code review report for your Python script. This report will cover code readability, naming conventions, formatting, error handling, documentation quality, and maintainability. Please provide the Python script when you're ready to proceed with the first step of the plan.

- **Receive Python Script:** Obtain the Python script from the user for review. This is the initial step before any analysis can begin.
- **Analyze Code Readability and Naming Conventions:** Examine the script for clarity, simplicity, and ease of understanding. This includes evaluating the use of descriptive names for variables, functions, and classes, and adherence to standard Python naming conventions.
- **Analyze Formatting, Style, and Documentation:** Evaluate the script's adherence to PEP 8 guidelines, consistent indentation, spacing, and other style aspects. Additionally, assess the quality and presence of docstrings for modules, classes, and functions, as well as inline comments for clarity and usefulness.
- **Analyze Error Handling and Maintainability:** Assess how errors and exceptions are managed within the script, including the use of try-except blocks and specific exception types. Evaluate the overall ease of modifying, extending, or debugging the code, considering factors like modularity and complexity.
- **Identify Quality Issues, Risk Areas, and Code Smells:** Pinpoint specific problems, potential risks, and common anti-patterns (code smells) within the codebase based on the previous analytical steps.
- **Generate Improvement Suggestions:** Formulate actionable recommendations to address the identified quality issues, mitigate risks, and eliminate code smells, aiming to improve overall code quality.
- **Compose AI-Based Automated Code Review Report:** Consolidate all findings, including identified quality issues, risk areas, code smells, and improvement suggestions, into a comprehensive and structured report.
- **Final Task:** Summarize the comprehensive AI-generated code review report and offer the option to provide an improved version of the code based on the report's suggestions.

Generating...

- Preparing
- Receive Python Script

What can I help you build?

Gemini 2.5 Flash

Gemini can make mistakes, so double-check it and use code with caution. Learn more

Variables Terminal

23:38 Python 3

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[4]

```python
print("Starting docstring analysis...")

# Parse the script into an AST
tree = ast.parse(python_script_content)

# 1. Module-level docstring
module_docstring = ast.get_docstring(tree)
print(f"\nModule-level docstring present: {'Yes' if module_docstring else 'No'}")
if module_docstring:
    print(f"  Docstring content (first 50 chars): {module_docstring[:50]}...")

# 2. Class and Function docstrings
class_info = []
function_info = []

for node in ast.walk(tree):
    if isinstance(node, ast.ClassDef):
        class_name = node.name
        class_docstring = ast.get_docstring(node)
        class_info.append(f"Class '{class_name}': {'Yes' if class_docstring else 'No'}")
    elif isinstance(node, (ast.FunctionDef, ast.AsyncFunctionDef)):
        function_name = node.name
        function_docstring = ast.get_docstring(node)
        function_info.append(f"Function '{function_name}': {'Yes' if function_docstring else 'No'}")

print("\n--- Class Docstring Analysis ---")
if class_info:
    for info in class_info:
        print(info)
else:
    print("No classes found.")

print("\n--- Function Docstring Analysis ---")
if function_info:
    for info in function_info:
        print(info)
else:
    print("No functions found.")

print("\nDocstring analysis complete.")
```

```
Starting docstring analysis...

Module-level docstring present: No

--- Class Docstring Analysis ---
Class 'MyClass': No

--- Function Docstring Analysis ---
Function 'my_func': No
Function '__init__': No
Function 'getdata': No

Docstring analysis complete.
```

Gemini

report will cover code readability, naming conventions, formatting, error handling, documentation quality, and maintainability. Please provide the Python script when you're ready to proceed with the first step of the plan.

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Generating...

- Preparing
- Receive Python Script
- Analyze Code Readability and Naming Conventions
- Analyze Formatting, Style, and Documentation
- Analyze Error Handling and Maintainability
- Identify Quality Issues, Risk Areas, and Code Smells
- Generate Improvement Suggestions

Show me the AI-generated code review report | Provide the improved version of the code | Perform a code review on a new Python script

What can I help you build?

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Gemini can make mistakes, so double-check it and use code with caution. Learn more

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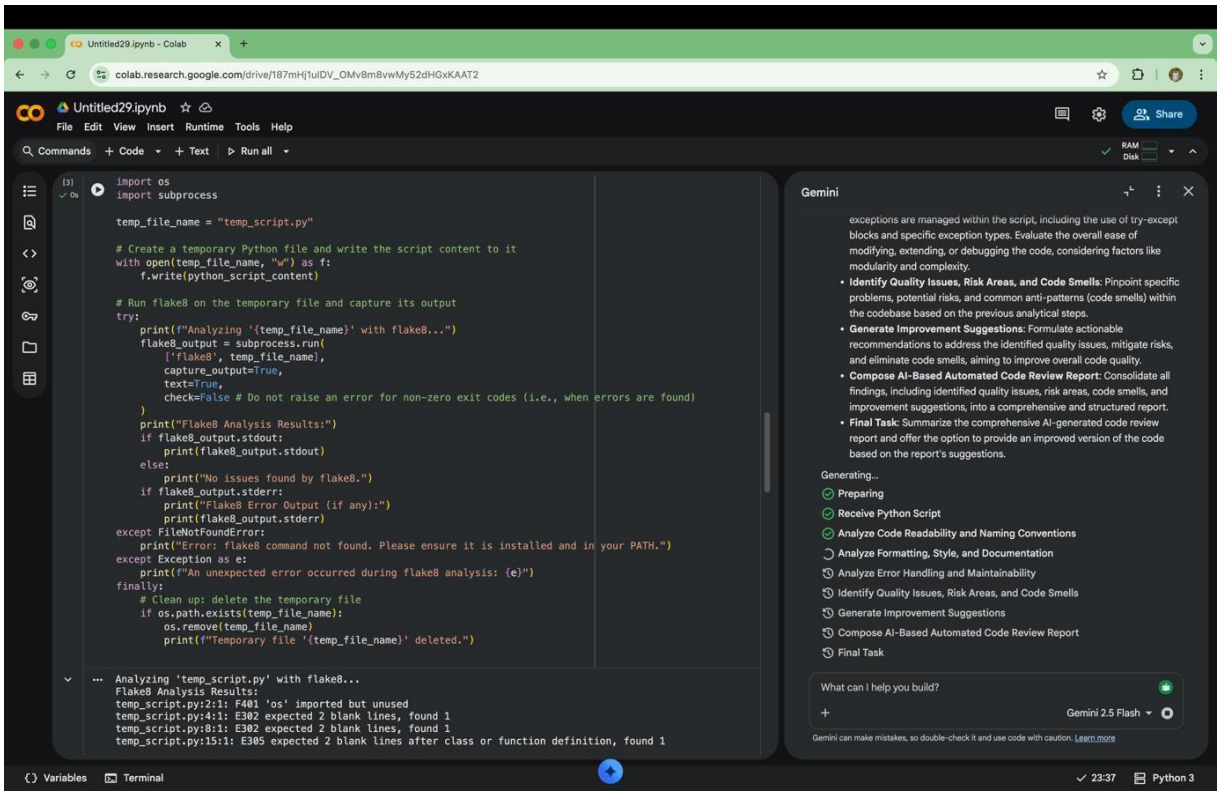

CODE EXPLANATION:

Documentation & Style: The script lacks docstrings and violates PEP 8 blank line conventions, hindering readability.

Naming & Imports: ARG_TWO uses an inconsistent naming convention, and the os module is imported but unused.

Error Handling: There is no explicit error handling, making the script fragile to unexpected inputs.

Maintainability Risk: These issues collectively reduce maintainability, increase cognitive load, and risk runtime crashes.

Improvements: Add docstrings, fix naming and formatting, remove unused imports, and implement robust error handling.